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SUPERSEDING
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FEDERAL SPECIFICATION

GRINDING MACHINES, VALVE FACE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers grinding machines used for reconditioning intake and exhaust valves of internal combustion engines by wet grinding.

1.2 Classification.

1.2.1 Types and styles. The grinding machines shall be of the following types and styles, as specified (see 6.2):

Type I - Light duty (1 to 4 inch valve head capacity).

Style A - For bench mounting.
Style B - Cabinet mounted.

Type II - Heavy duty, floor mounted (1 to 5-3/8 inch valve head capacity).

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications

CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)

Federal Standards

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-123 - Marking for Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly

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supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outisde the Federal Government for bidding purposes, are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-M-18058	-	Machinery, Metal and Woodworking, Packaging of

Military Standards

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulation:

Occupational Health and Safety Administration (OSHA) 29 CFR 1910.211-222 General Industrial Safety and Health Standards

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Gear Manufacturers' Association (AGMA)

AGMA 360.02	-	Manual for Machine Tool Gearing
AGMA 390.03	-	Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

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American National Standards Institute (ANSI)

ANSI B7.1 - Safety Requirements For The Use, Care, and Protection of Abrasive Wheels
ANSI B46.1 - Surface Texture
ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives

(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging
ASTM E380-84 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

International Organization for Standardization (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Fire Protection Association (NFPA)

NFPA 70 - National Electrical Code
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one or more complete grinding machines for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

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3.2 Design. The grinding machine shall be designed specifically for regrinding the faces of intake and exhaust valves of internal combustion engines and for the related operations of valve butt and push rod grinding, rocker arm grinding, and the grinding of tappets and adjustable screw faces. The machine shall be designed for traverse grinding, as opposed to plunge grinding. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair without modification, distortion, or damage to the machine or any machine part or component.

3.2.1 Maintainability. Design and assembly of the grinding machine and the placement of adjusting features shall insure that operational maintenance capability will be reduced to the simplest form of procedures requiring a minimum of skills and common tools. Instructions for disassembly, repair, assembly, and adjustment shall be clear, concise, and definitive in application.

3.2.2 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.2.3 Safety devices. Safety devices shall be provided for parts that present safety hazards to personnel. Safety devices shall not interfere with the operation of the machine or with the operator's performance. The operator's manual shall clearly outline any safety precautions relevant to the assembly, operation, disassembly, and repair of the machines.

3.2.3.1 Occupational safety standards. The product shall comply with OSHA standards set forth in 29 CFR 1910.211-222 that are applicable to grinding machines.

3.2.4 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.5 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the grinding machines. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.6 Measuring and indicating devices. When specified (see 6.2), measuring and indicating devices shall be graduated in both the US and SI systems of measurement. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the grinder shall be graduated in the same system.

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3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The grinding machine shall be complete so that it can be used for operations specified herein.

3.4.1 Castings and forgings. All castings shall be free from blow-holes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclosure, cold shuts, excessive mismatching, and other defects that may affect their structural strength. The casting's strength, and other essential properties, shall be adequate to meet the performance requirements specified herein.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise deformed.

3.4.4 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.4.5 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

3.4.6 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

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3.4.7 Gears. All gears and pinion designs selected for use in the grinding machine or its components shall be fabricated to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and 1122/1 and ASTM E380.

3.5 Performance and product characteristics.

3.5.1 Performance. The grinding machine shall be capable of withstanding without damage, malfunction, or loss of accuracy continuous operation for the period and production rate specified herein. Steel and stellite valves (used) with face angles of 30° and 45°, head diameters of 1-1/2 to 2 inches, and stem diameters of 5/16 to 3/8 inch shall be used for refacing. (For the purpose of determining conformance to the requirements of this specification, the term "refacing" shall be defined as the removal of the amount of stock necessary to completely resurface the face of the valve or the removal of at least 0.003 inch of stock from the circumference and across the width of the valve face, whichever is greater.)

3.5.2 Types and styles. The type I, style A, grinding machine shall be dimensionally suitable for mounting on a work bench 28 to 32 inches high. The type I, style B, grinding machine shall embody the same characteristics as style A except that it shall be mounted on a cabinet, as specified herein. The type II grinding machine shall be mounted on a cast iron or fabricated steel floor-type mounting.

3.5.3 Accuracy. The grinding machine shall be capable of grinding valve faces to within one degree less than the indicated angle and shall provide a grinding accuracy of one half degree per setting. The surface roughness of the ground valve shall not exceed 32 microinches, arithmetical average (aa), as determined in accordance with ANSI B46.1. When measured on a test mandrel at least one inch from the nose of the chuck, concentricity of the spindle, workhead, and chuck shall not exceed 0.001 inch total indicator reading (tir).

3.5.4 Capacity. The grinding machine shall be capable of grinding valves with stem and head diameters in accordance with the following information:

Type	Stem diameter (inches)	Head diameter (inches)
I	9/32 to 11/16	1 to 4
II	9/32 to 1-1/4	1 to 5-3/8

3.5.4.1 Angular capacity. The grinding machine shall be capable of grinding valve faces to any angle from 15° to 45°. The angle setting device shall be marked to enable the accurate positioning and locking of the machine components for the grinding of valve faces at angles of 15°, 29°, 30°, 44°, and 45°. The acute angles, so defined, are subtended between the ground valve face and a line or surface perpendicular to the valve stem centerline.

3.5.5 Endurance and production rate.

3.5.5.1 Type I. The type I grinding machine shall be capable of performing, as specified in 3.5.1, for a period of at least two hours while maintaining a production rate, during that period, of at least 20 valves per hour.

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3.5.5.2 Type II. The type II grinding machine shall be capable of performing, as specified in 3.5.1, for a period of at least two hours while maintaining a production rate, during that period, of at least 40 valves per hour.

3.5.6 Operation. The movement of the workhead, necessary in traverse grinding, shall be controlled by means of a ball crank, handwheel, or lever. The grinding feed shall be controlled by a feed screw actuated by a ball crank or handwheel. The feed shall be indicated in increments of 0.001 inch, with numerical values at intervals not exceeding 0.010 inch. When a power operated chuck is furnished, means shall be provided for temporary conversion to manual operation. The workhead spindle and chuck speed shall be adjustable to the extent necessary to enable the refacing of all valves within the applicable size range specified in 3.5.4.

3.6 Details of components. The grinding machine shall consist of, but not be limited to, the following items.

3.6.1 Cabinet. The type I, style B grinding machine shall be mounted on a closed cabinet made of reinforced sheet steel not less than Manufacturers Standard Gage No. 20. The cabinet shall contain drawers or compartments for storage of spare parts and accessories. The cabinet shall be capable of withstanding, without damage, a balanced load equal to at least twice the weight of the grinding machine. Dimensions of the cabinet shall be such that the grinding wheel spindle is 40 to 46 inches above floor level. When specified (see 6.2), the cabinet shall be mounted on four rubber tired casters at least three inches in diameter. At least two of the casters shall be provided with footlocks for holding the cabinet stationary during operation of the grinding machine.

3.6.2 Mounting, type II. The grinding machine shall be furnished with a metal base for floor mounting. The mounting, furnished with the grinding machine, shall contain at least three bolt holes to facilitate permanent installation of the unit. The mounting shall have sufficient strength, weight, rigidity, and bearing area to support the grinding machine and to prevent vibration marks from being transferred to the work. The mounting shall be of such height that the grinding wheel spindle shall be from 40 to 46 inches above floor level. The mounting may be cabinet or pedestal style.

3.6.3 Base. The base of the grinder shall be made of an iron casting or steel weldment. The base shall support and secure alignment of the wheelhead and workhead assemblies.

3.6.4 Wheelhead assembly. The housing of the wheelhead assembly shall be assembled to the base of the machine in a manner to provide the accuracy and stability necessary to meet the requirements of this specification.

3.6.5 Workhead assembly. The housing of the workhead assembly shall be assembled to the base in a manner to enable the operator to swivel the workhead to the extent necessary to meet the valve face grinding requirements specified herein.

3.6.6 Spindle bearings. The wheelhead and workhead spindles shall be mounted on replaceable, precision-type antifriction bearings sealed against abrasive dust and abrasive-laden coolant. Bearings shall be preloaded to prevent end play or provision shall be made to take-up end play.

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3.6.7 Chuck(s). The chuck(s) shall be of the self-centering, quick-opening, quick-closing type and shall provide accuracy and capacity within the limits specified herein. The gripping and releasing action shall be controlled by a handwheel, wrench, or lever. Surfaces of the chuck(s) coming in contact with the valve stem shall be hardened to 55 to 60 Rockwell C.

3.6.8 Guides and ways. Guides and ways, used for the support and alignment of the grinding machine components, shall be provided with wipers and a means of adjustment to compensate for wear when such adjustment is necessary. The surface roughness of round or tubular ways and guides shall not exceed 32 microinches aa. When dovetail ways and guides are used, surface roughness shall not exceed 63 microinches aa. Surface roughness shall be determined as specified in ANSI B46.1.

3.6.9 Coolant system. Each grinding machine shall be provided with a coolant system which shall be an integral part of the machine. The coolant system shall consist of a storage tank with baffles for the separation and settlement of chips and abrasive particles, a pump, and all piping, nozzles, valves, and channels necessary to deliver and adjust the flow of coolant to the points of contact between the work and the grinding wheel(s) and to return the coolant to the storage tank. The storage tank shall be accessible for cleaning.

3.6.9.1 Coolant tank capacity. The capacity of the coolant tank and the efficiency of the circulating system and splash guards shall be sufficient to provide coolant for continuous operation, without augmenting the coolant supply, during the periods specified in 3.5.5.1 and 3.5.5.2.

3.6.10 Grinding wheel(s). The grinding machine shall be provided with two grinding wheels; one wheel for grinding stellite valve faces and one wheel for chamfering and squaring valve stem ends. Wheel hoods or guards shall be provided. Guards shall also be provided on all belt drives. The wheels shall conform to ANSI B7.1 and B74.13. The grinding wheel characteristics in relation to the grinding wheel spindle speed shall be such that the wheel or wheels shall continue to meet the performance requirements of this specification after the original diameter has been reduced 15 percent by dressing.

3.6.11 Grinding wheel dresser. The grinding machine shall be equipped with a grinding wheel dressing fixture and a diamond dressing stone designed to perform all necessary truing and dressing operations. The dressing stone for the type I grinding machine shall be at least 0.15 carat in weight. The dressing stone for the type II grinding machine shall be commensurate with the best quality in the grinding tool industry, shall have a minimum of four good cutting points, and shall be not less than 0.25 carat in weight.

3.6.12 Rocker-arm grinding attachment. A rocker-arm grinding attachment shall be provided with each grinding machine. The attachment shall be designed to facilitate refinishing the radii of the rocker arms of all engines having valves within the capacity range of the grinding machine.

3.6.13 Valve stem grinding attachment. A valve stem grinding attachment shall be provided to enable the grinding of valve butts, push rods, tappets, and adjusting screw faces. The attachment shall be so arranged that valve stem ends can be ground square, as indexed from the ground face of the valve, to a predetermined length within a tolerance of $\pm$0.0005 inch. The feed of the valve

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stem grinding attachment shall be controlled by a knurled knob or handwheel. The feed adjustment shall be indicated in increments of 0.001 inch with numerical values, at intervals, not exceeding 0.010 inch.

3.6.14 Depth gage. A depth gage shall be furnished with each grinding machine. The depth gage shall be designed to measure the distance from the valve seat to the push rod, and to transfer the measurement to the valve stem grinding attachment.

3.6.15 Electrical equipment. Unless otherwise specified (see 6.2), electrical components shall conform to, and shall be located in accordance with, NFPA 79.

3.6.15.1 Motors. All motors shall be continuous duty type in accordance with CC-M-1807.

3.6.15.1.1 Grinding wheel drive motor. The grinding wheel drive motor for types I and II grinding machines shall have ball bearings and continuous duty rating. Unless otherwise specified (see 6.2), the motor for the type I grinding machine shall be capable of operation on nominal 115 volts, 60 hertz (Hz), single phase current. The motor for the type II grinding machine shall be capable of operation on nominal 230 volts, 60 Hz single phase current. Splash guards or other means shall be provided to preclude entry of coolant into the motor.

3.6.15.1.2 Workhead drive and pump motors. When separate motors are used for workhead or coolant pump drive, they shall be capable of operation on the same voltage, frequency, and phase as the grinding wheel drive motor.

3.6.15.2 Switches. For the type I grinding machine, the start and stop switch for control of the grinding wheel drive motor shall be either a pushbutton or toggle type. For the type II grinding machine, the start and stop switch shall be of the pushbutton type. If the workhead is driven by a separate motor, an independent switch shall be furnished for its control. The switches shall be located in a manner to be accessible from the position normally occupied by the operator.

3.6.15.3 Power supply cord. Each machine shall be furnished with a power supply cord properly wired into the machine. The cord shall have a voltage and current carrying capacity equal to, or more than that required by, the motor(s) and NFPA 70, National Electric Code. The power supply cord shall be at least 10 feet long and terminate in one inch of stripped, tinned wire without a plug for a terminal.

3.6.15.4 Terminal connector. All motor and motor control wiring on the type II grinding machine shall be connected to a terminal box on the machine to facilitate installation.

3.6.15.5 Light fixture. The grinding machine shall be equipped with a shaded work light. The work light position shall be adjustable to the extent necessary to provide illumination for all grinding operations.

3.6.15.6 Dielectric strength. The machine electrical system shall withstand without damage a root mean square (rms) voltage of 1,000, plus twice its rated voltage, applied for a period of one minute between mutually insulated circuits

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and between insulated circuits and ground. The voltage shall be at a frequency of 60 Hz.

3.7 Product identification. The grinding machine shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.8 Technical data. Data shall be furnished as specified (see 6.3 and 6.3.1).

3.9 Workmanship. Standards of workmanship shall assure that the grinding machine shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production units offered for acceptance under the contract and shall consist of (a) through (f) below. Failure of any unit to pass an examination, test, or inspection shall be cause for rejection of the unit.

(a) Examination (see 4.3).
(b) Chuck concentricity test (see 4.4.2.1).
(c) Performance test, type I or II (see 4.4.2.2 or 4.4.2.3) (except that the test shall consist of grinding only the smallest and the largest valve (one each) within the grinding machine capacity range to the accuracy requirements of 3.5.3).
(d) Valve stem grinding test (see 4.4.3).
(e) Dielectric test (see 4.4.4).
(f) Packaging inspection (see 4.5).

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4.2.2.1 Sampling. Sampling for quality conformance inspection shall be as follows:

(a) Examination (see 4.3): Each production unit.
(b) Chuck concentricity test (see 4.4.2.1): Each production unit.
(c) Performance test (see 4.4.2.2 and 4.4.2.3): Each production unit.
(d) Valve stem grinding test (see 4.4.3): Each production unit.
(e) Dielectric test (see 4.4.4): One production unit.
(f) Packaging inspection (see 4.5): General Inspection Level II and an Acceptable Quality Level (AQL) of 4.0 percent defective in accordance with MIL-STD-105.

4.3 Examination. The grinding machines shall be visually and manually examined to determine conformance with the requirements of Section 3. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the item has passed the examinations prescribed by the contractor's inspection system shall be cause for rejection.

4.3.1 Evidence of compliance with OSHA standards. Evidence of compliance with applicable safety and health requirements shall be reviewed (see 3.2.3.1 and 6.4).

4.4 Tests.

4.4.1 Test conditions. All tests shall be conducted while the grinding machine is being operated in accordance with the technical data specified in 3.8. The ambient temperature shall be 70° to 80°F. At least 50 percent of the valves ground during the tests shall be made of stellite. The remainder shall be made of steel.

4.4.2 Test methods.

4.4.2.1 Chuck concentricity test. A test mandrel 9/32 inch in diameter shall be placed in the grinding machine chuck. The workhead spindle shall be rotated and an indicator reading shall be taken on the test mandrel at a distance of at least one inch from the nose of the chuck. The test shall be repeated on one mandrel with a diameter equal to the maximum chuck capacity and on two mandrels of intermediate sizes. The test shall be performed on each mandrel three times, removing and reinstalling the mandrel for each test. If the tir exceeds 0.001 inch in any case, the grinding machine shall be rejected (see 3.5.3).

4.4.2.2 Type I grinding machines. The performance test for the type I grinding machine shall be as follows. The grinding machine shall be operated continuously for a period of at least one hour under the following conditions:

(a) The grinding machine shall be used to reface valves with head diameters of 1-1/2 to 2 inches and stem diameters of 5/16 to 3/8 inch, at a production rate of at least ten valves per half hour.

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(b) The angle setting device shall be positioned, alternately, every twenty valves at 29° for the 30° valves and at 44° for the 45° valves.

One valve shall be selected for inspection from the last five valves ground during each hour of operation. If the valve face angle is not within plus 0 to minus 10 minutes of the predetermined angle, or if the surface roughness of the valve face exceeds 32 microinches aa, the grinding machine shall be rejected (see 3.5.1, 3.5.3, and 3.5.5.1).

4.4.2.3 Type II grinding machines. The performance test for the type II grinding machine shall be the same as that specified in 4.4.2.2, except that the production rate shall be at least 40 valves per hour and the angle setting device shall be positioned alternately at 30° and 45° every 40 valves (see 3.5.1, 3.5.3, and 3.5.5.2).

4.4.3 Valve stem grinding test. A valve, the overall length of which has been determined previously, shall be placed in the valve stem grinding attachment and fed toward the wheel until the end of the valve contacts the grinding wheel. The feed control shall then be turned to the extent necessary to indicate the removal of 0.010 inch of stock from the end of the stem. The valve shall then be removed from the attachment and the overall length shall be measured again to determine the amount of stock actually removed. If the reduction in overall length is not within $\pm$0.0005 inch of the indicated 0.010 inch, the grinding machine shall be rejected (see 3.6.13).

4.4.4 Dielectric test. The wiring of the electrical system, including motor(s) and switches, shall be subjected to a dielectric test of a voltage equal to twice the rated voltage plus 1,000 rms (2X + 1,000) for a period of one minute between the specified circuit and the grinding machine housing. After completion of test, the machine and components shall be examined for damage (see 3.6.15.6).

4.5 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Military preservation, packing, and marking. When specified (see 6.2), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-M-18058.

5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

5.3 Marking for civil agencies. When specified (see 6.2), marking for shipment for civil agencies shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The grinding machines covered by this specification are intended for use in reconditioning intake and exhaust valves of internal combustion engines. The type I, style A, grinding machine is designed for bench mounting, and has a valve stem capacity of 9/32 to 11/16 inch and a valve head

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capacity of one to four inches. The type I, style B, grinding machine is the same as style A except that it is mounted on a metal cabinet. The type II grinding machine is a heavy duty, floor-mounted unit capable of grinding valves with stem diameters of 9/32 to 1-1/4 inches and head diameters of 1 to 5-3/8 inches.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and style required (see 1.2.1).
(c) First article when required (see 3.1).
(d) Measuring and indicating device graduation(s) required (see 3.2.6).
(e) Painting and finishing if different (see 3.4.4).
(f) Rubber tired casters when required (see 3.6.1).
(g) Electrical equipment if different (see 3.6.15).
(h) Voltage, frequency, and phase for grinding wheel drive motor if different (see 3.6.15.1.1).
(i) Product identification if different (see 3.7).
(j) First article inspection if different (see 4.2.1).
(k) Quality conformance inspection if different (see 4.2.2).
(l) Level of preservation, packing, and marking if different (see 5.1 and 5.2).
(m) Civil agency shipment marking when required (see 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract (see 6.3.1).

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.4 OSHA standards. Products which meet all requirements of this specification comply with the OSHA standards published prior to the date of this specification and identified in Section 3 (see 3.2.3.1).

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00-G-686C

MILITARY INTERESTS:

Custodians
Army - AL
Navy - YD
Air Force - 99

Review Activities
Army - ME
Air Force - 82

User Activity
Army - AT

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army - AL

DoD Project 4910-0583

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